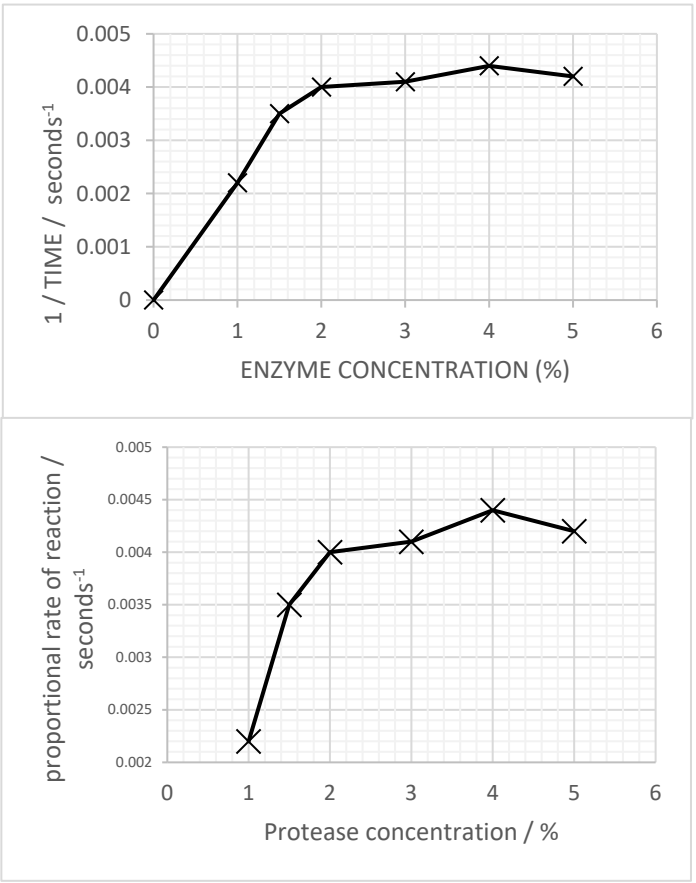


Question Number	Answer	Additional Guidance	Mark
3(a)(i)	time taken for the film to become clear	accept how long it takes	1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	An description including the following points: • carry out the experiment described at constant {temperature / substrate / enzyme concentration} (1) • at a range of pHs (controlled by buffers) (1) • choose the pH which gives a time which is not {too short / too long} (for practicality) (1)		3

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	A graph with the following features: • A axes correctly orientated (1) • L axes correctly labelled and with correct units (1) • P correct plotting on a linear scale (1) • R points joined with ruled straight line (1)	E.G.  The top graph shows a plot of $1 / \text{TIME} / \text{seconds}^{-1}$ on the y-axis (ranging from 0 to 0.005) against ENZYME CONCENTRATION (%) on the x-axis (ranging from 0 to 6). The data points are connected by a ruled straight line, showing a rapid increase in the reciprocal of time as enzyme concentration increases, eventually leveling off. The bottom graph shows a plot of proportional rate of reaction / seconds^{-1} on the y-axis (ranging from 0.002 to 0.005) against Protease concentration / % on the x-axis (ranging from 0 to 6). The data points are connected by a ruled straight line, showing a rapid increase in the proportional rate of reaction as protease concentration increases, eventually leveling off.	4

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	An answer including the following points: • as enzyme concentration increases (proportional) rate of reaction also increases (1) • as enzyme concentration increases there are more {active sites / collisions / ES complexes formed} (1) • it levels off (1) • because substrate becomes limiting (1)	accept time taken decreases	4

Question Number	Answer	Additional Guidance	Mark
3(c)(i)	An answer including the following points: • {the rate of the reaction slows down / gelatine is used up (over time)} (1) • so rates can be validly compared / compared only if they are all initial rates (1)	accept rate of reaction changes over time	2

Question Number	Answer	Additional Guidance	Mark
3(c)(ii)	An answer including the following points: • measure transparency of film over time (1) • plot transparency against time (1) • calculate (initial rate from) gradient of straight part of graph at the beginning (1)	accept measure intensity of colour of liquid in which film suspended accept plot of whatever measured for mp1 against time tangent	3

Question Number	Answer	Additional Guidance	Mark
3(d)	An answer including the following points: • Dilution factor calculated (1) • Volume of 5% solution and water stated (1)	e.g. $5 \div 2 = 2.5$ accept $C_1V_1 = C_2V_2$ accept 4 stock to 6 water e.g. 1 cm³ of 5% added to 1.5 cm³ of water 0.4 cm³ of 5% to 0.6 cm³ water 2 cm³ to 3 cm³ of water correct answer with no working gains 2 marks	2